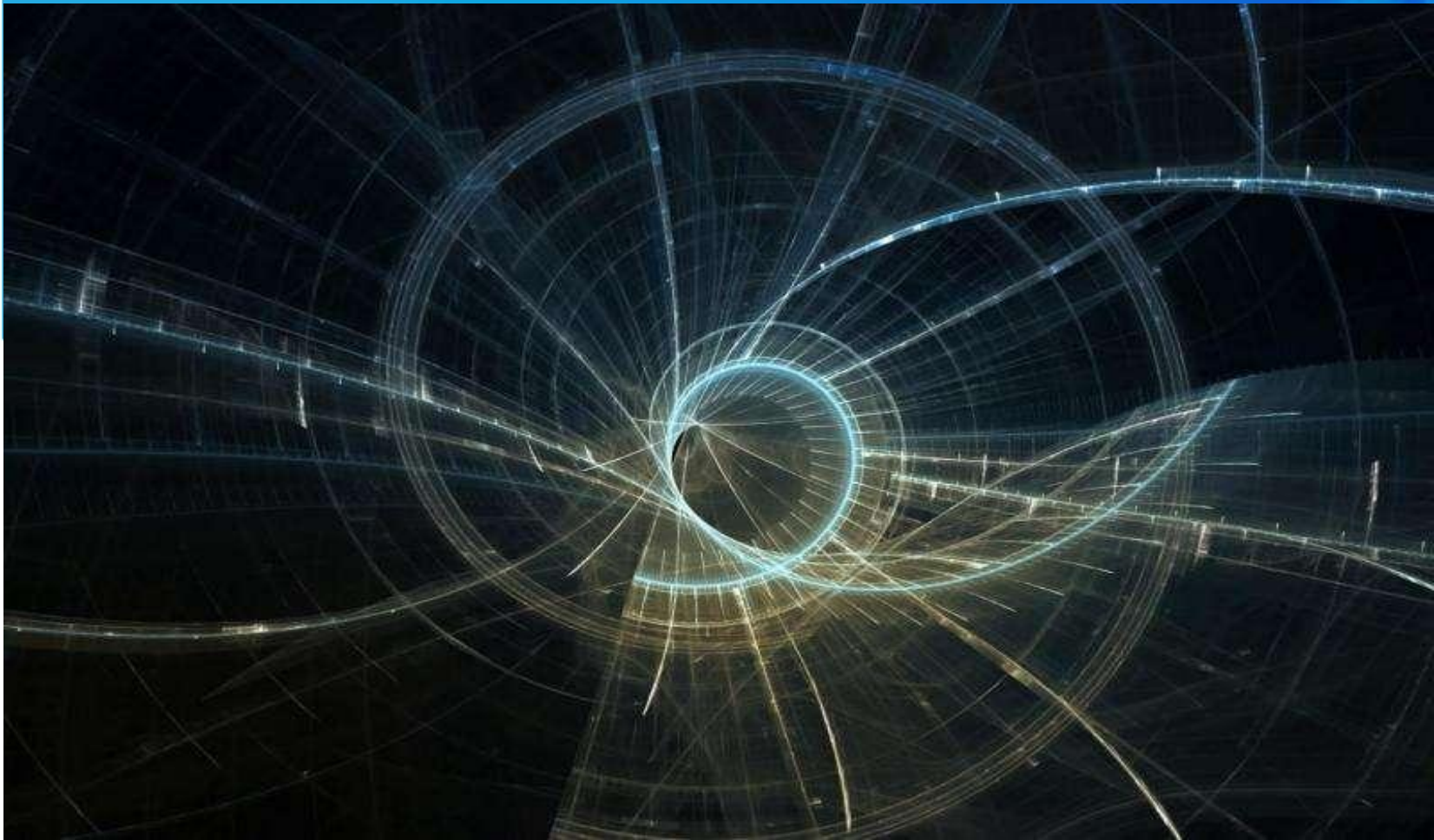


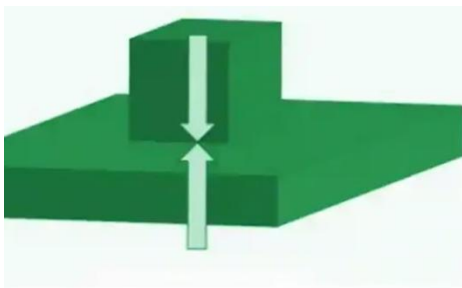
MECHANICS

Physics is the branch of science that explores the physical nature of matter and energy.



Mechanics: the study of how bodies react to forces acting on them

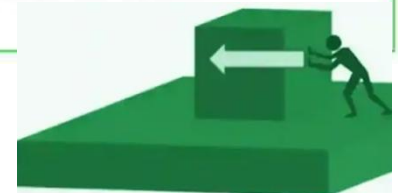
Statics: the study of bodies in equilibrium



Statics

Dynamics:

1. Kinematics – concerned with the geometric aspects of motion
2. Kinetics - concerned with the forces causing the motion



Dynamics

Mechanics in physics is the study of the motions of material bodies.

OR

Mechanics is the study of certain general relations that describe the interactions of material bodies and the Laws governing them.

It may be further subdivided into three sub-disciplines:

- **Dynamics**:- It is the study of the laws which determine, among all possible motions, which motion will actually take place in any given case.
 - **Kinematics**:- Study and description of the possible motions of material bodies.
 - **Kinetics**:- Study and description of forces causing motion of material bodies.
- **Statics**:- It is the study of forces and system of forces, with particular reference to the system of forces which act on bodies at rest.

- **DISPLACEMENT:** Change in position in specific direction.

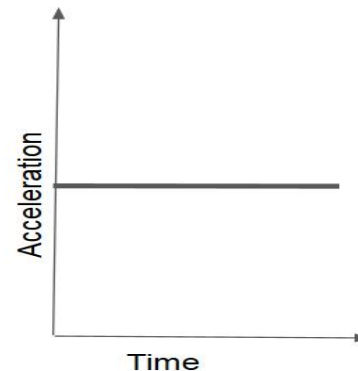
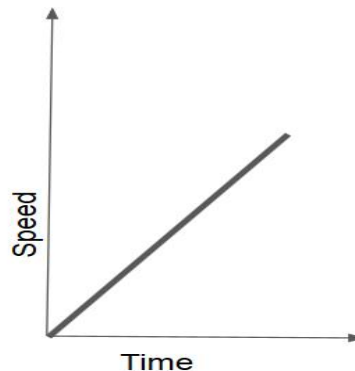
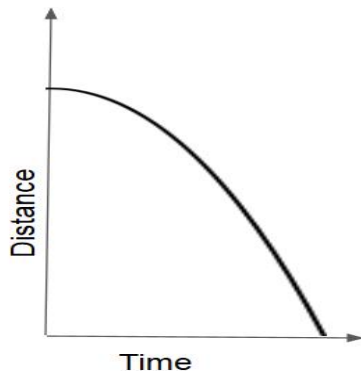
$$\Delta r = r_2 - r_1$$

- **VELOCITY:** Rate of change of position.

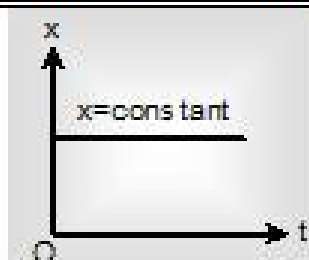
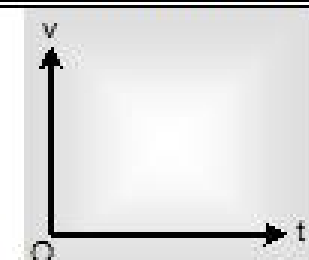
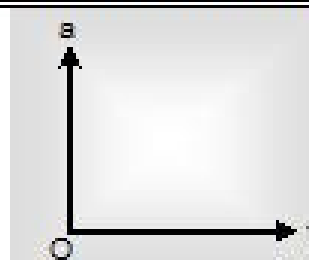
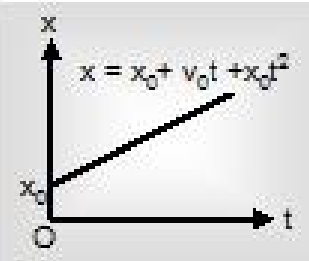
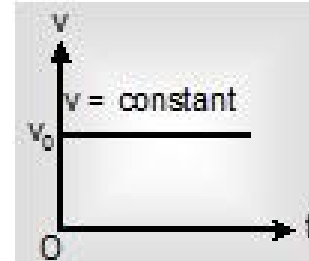
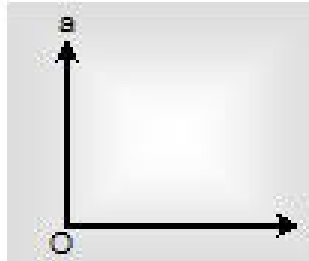
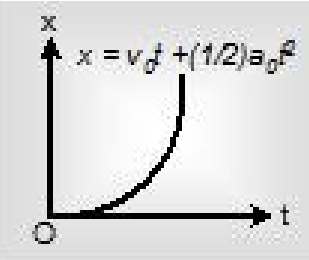
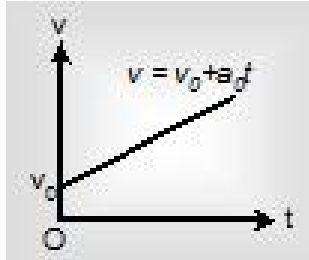
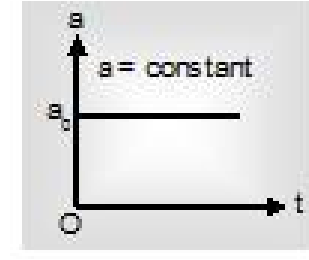
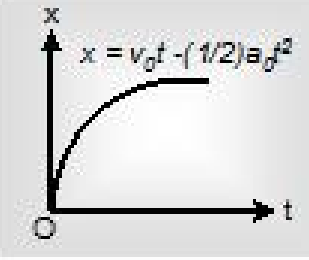
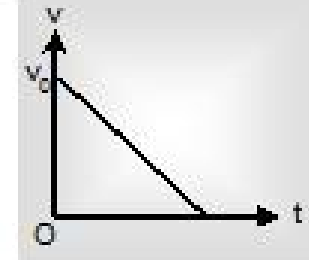
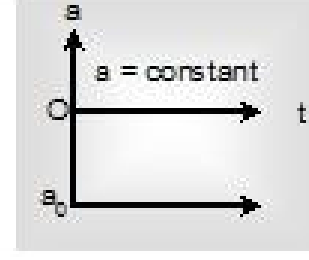
$$\Delta V = \frac{\Delta r}{\Delta t}$$

- **ACCELERATION:** Rate of change of velocity.

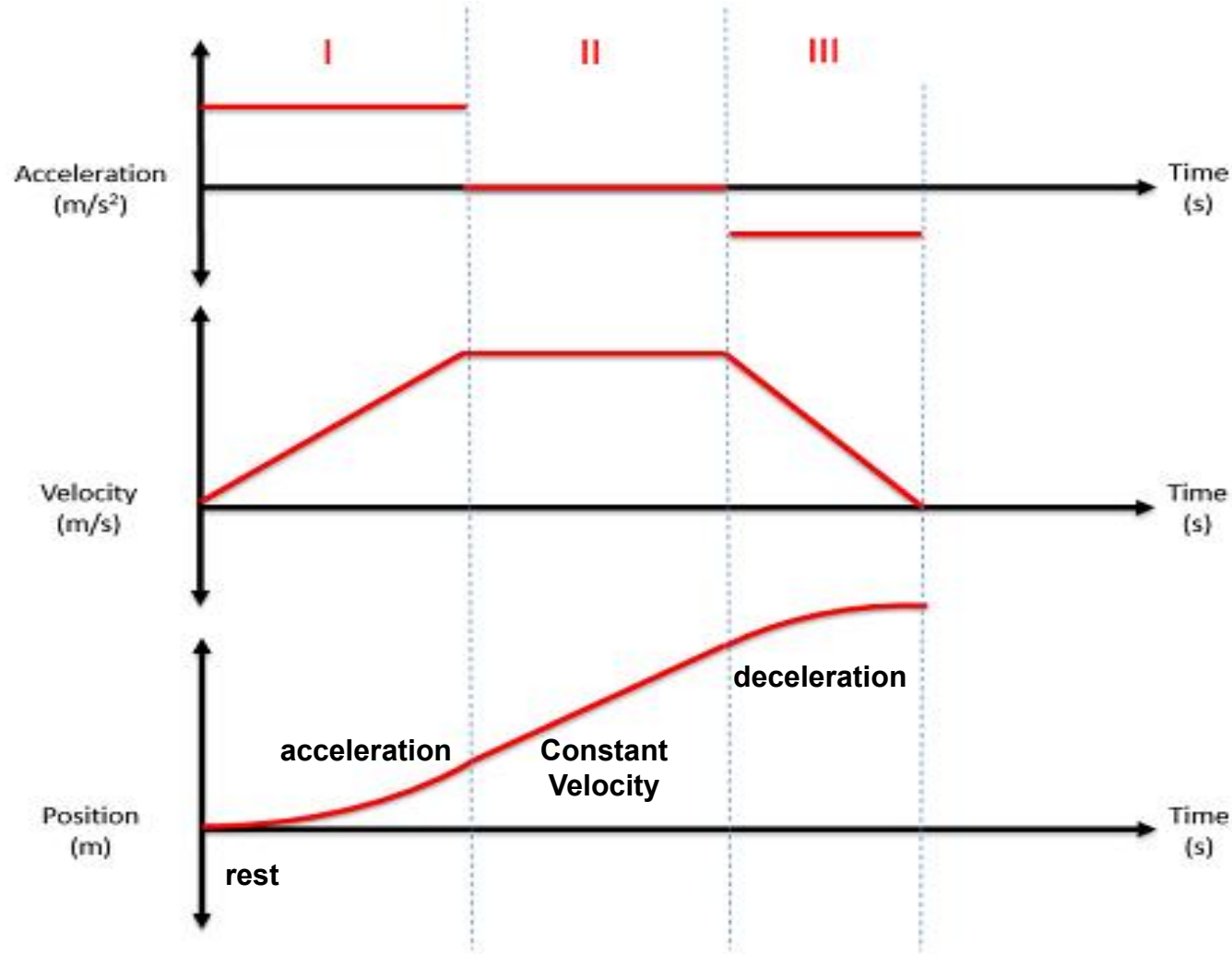
$$a_{avg} = \frac{\Delta V}{\Delta t}$$



Cases of motion:

	Displacement(x)	Velocity(v)	Acceleration (a)
a. At $v=0$;			
b. Motion with constant velocity			
c. Motion with constant acceleration			
d. Motion with constant deceleration			

e. Accelerating and braking car:



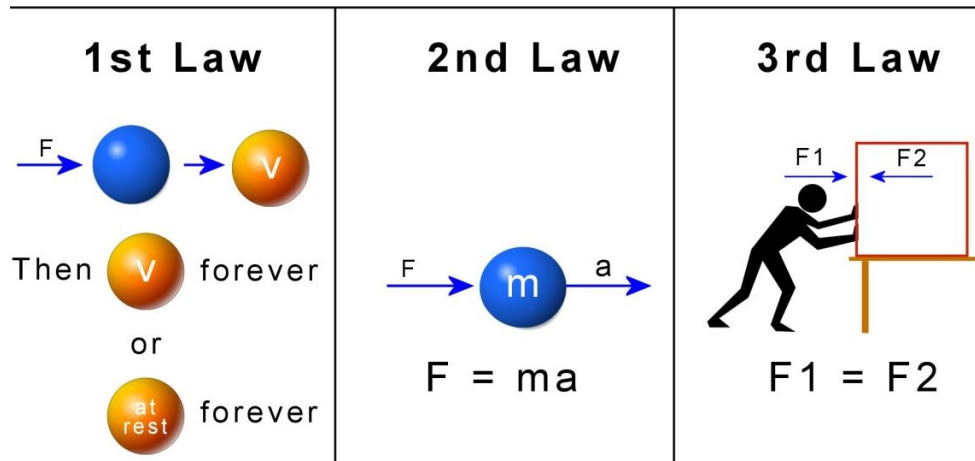
Newton's Law:

1st : If a body is at rest or moving at a constant speed in a straight line, it will remain at rest or keep moving in a straight line at constant speed unless it is acted upon by a force.

2nd : The time rate of change of the momentum of a body is equal in both magnitude and direction to the force imposed on it. Conversely, if a body is not accelerated, there is no net force acting on it. **$F = ma$**

(The momentum of a body is equal to the product of its mass and its velocity.)

3rd : When two bodies interact, they apply forces to one another that are equal in magnitude and opposite in direction.

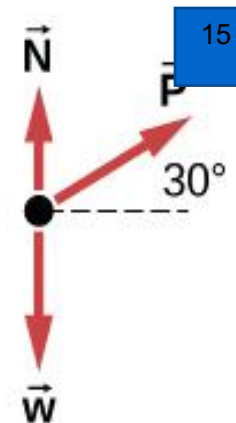
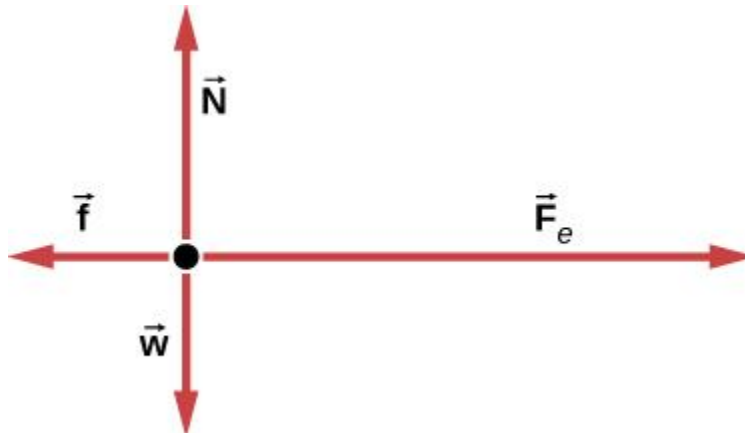
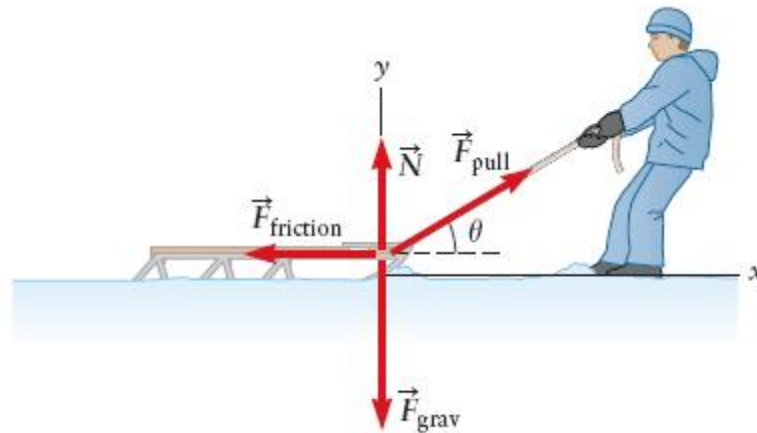


The Normal force:

- The force exerted by a surface on the object placed on it to cancel its $W=mg$.
- Always perpendicular (normal) to the surface.
- If the object (considering book placed on a table) is pushed downwards with a force P the book will remain at rest.
- So $a = 0$, the net force will also be 0.
- The total upward (normal) force will be,

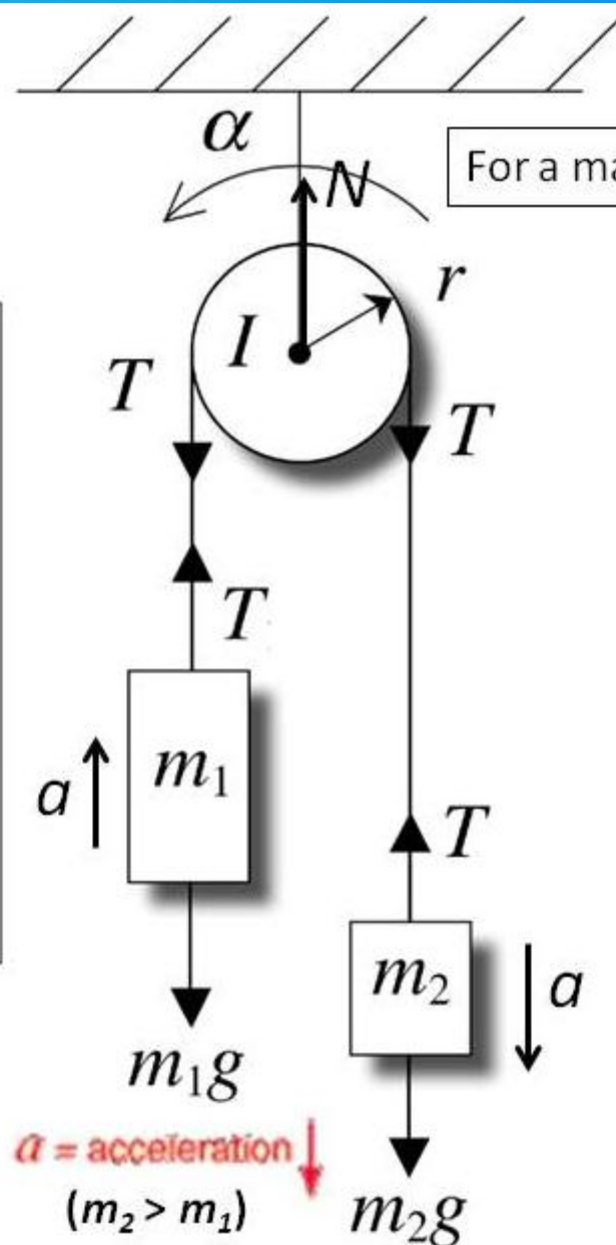
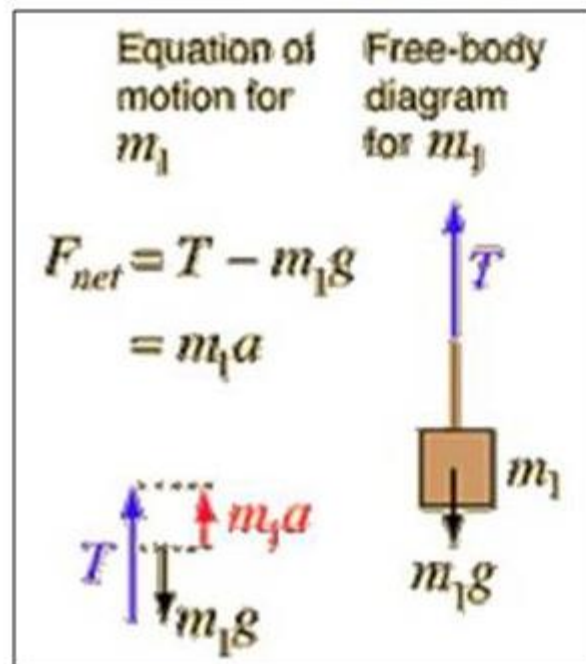
$$\mathbf{N = W + P}$$

A sled of mass $m = 7.5 \text{ kg}$ is pulled along a horizontal surface by a cord. A constant force $P = 21.0 \text{ N}$ is applied to the cord. Analyze the motion if,
 a) the cord is horizontal to the surface ($\theta = 0$), b) the cord makes an angle ($\theta = 15^\circ$) with the horizontal.

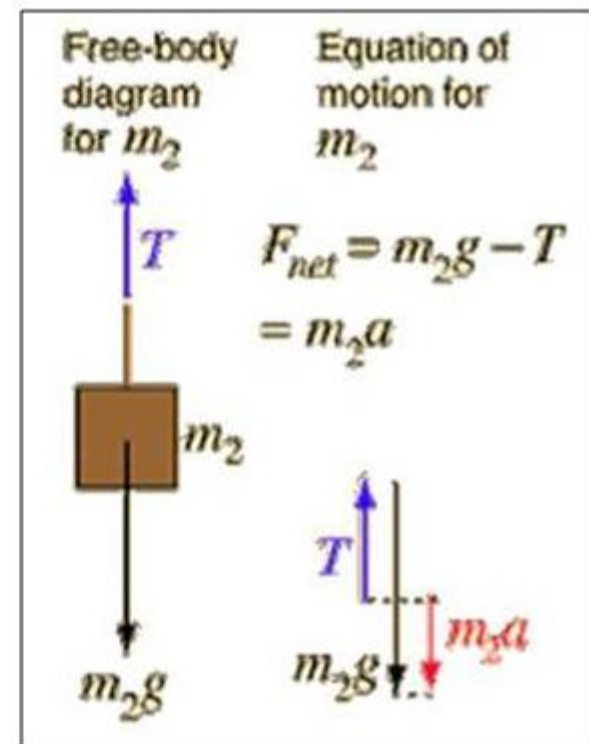


Tension in the string:

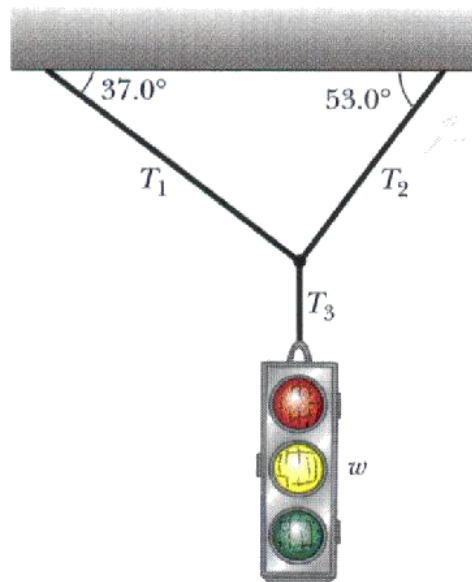
- **Case 01:** Two objects of unequal masses are attached to a frictionless Atwood machine having negligible mass of pulley. Determine the magnitude of acceleration of the two objects and tension in the light weight string.

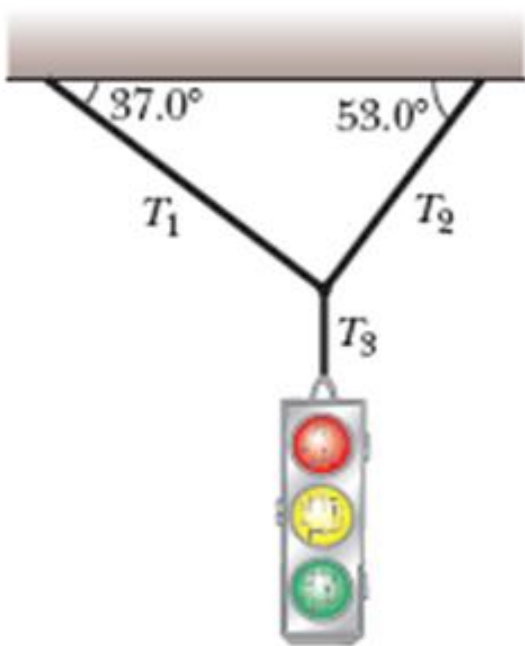


For a massless pulley: $N = T + T = 2T$



- **Case 02:** A traffic light having a weight of 122N is hanging from a cable attached to two other cables fastened to a support. The upper cable makes an $\theta = 37^\circ$ and $\theta = 53^\circ$ with the horizontal. The upper cables are not as strong as vertical cable and will break if tension exceed 100N.
- Will the traffic light remain hanging?

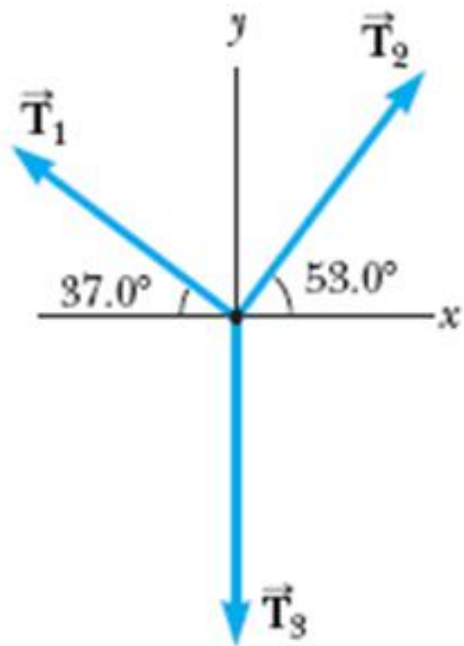




(a)



(b)



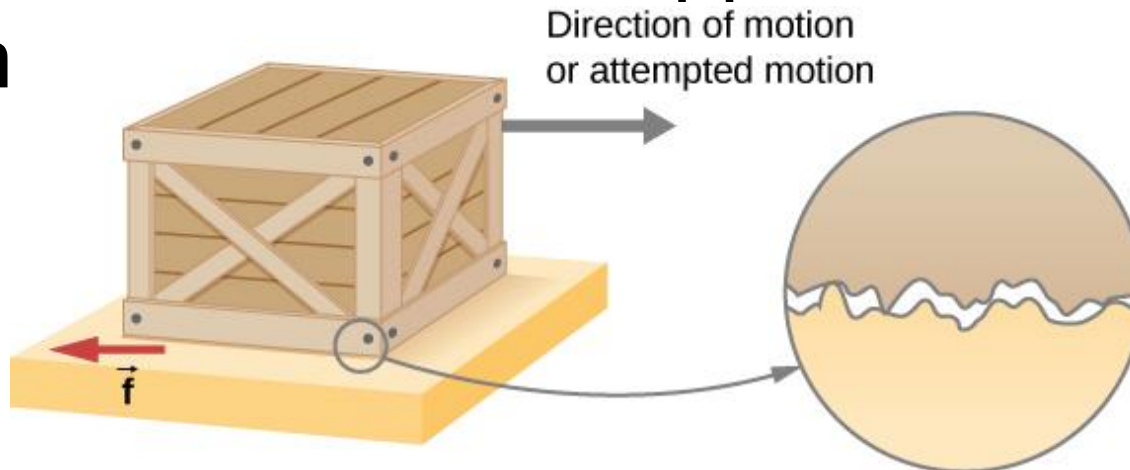
(c)

Frictional Forces:

“When an object is moving either on a surface or in a viscous medium such that there is resistance in its motion, the resistive force is called *frictional force*”.

OR

“Friction is a force that opposes relative motion

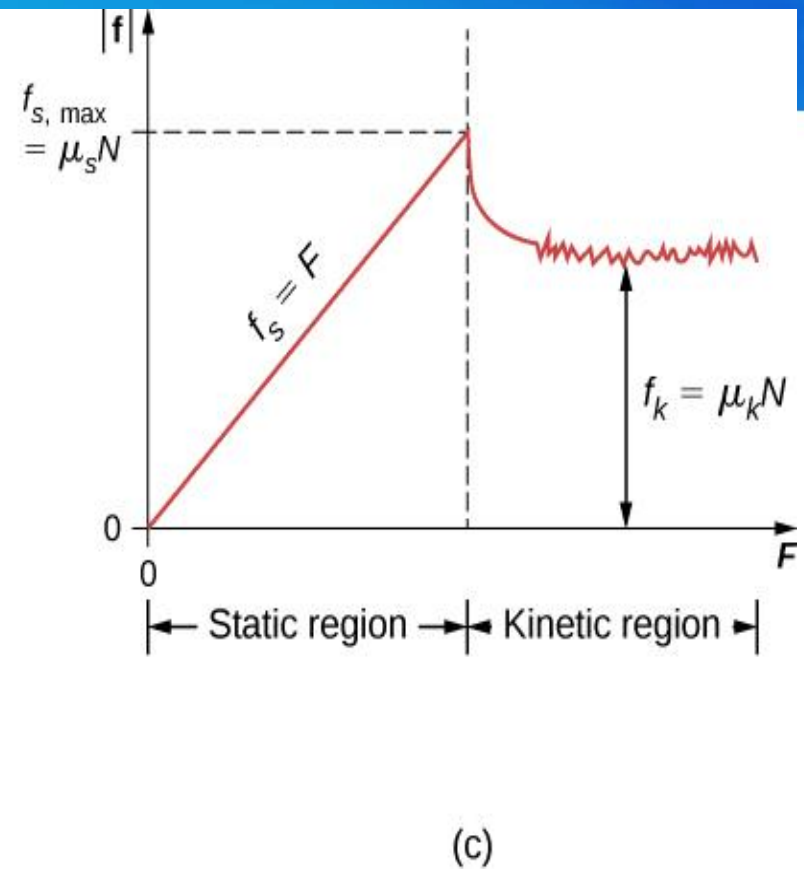
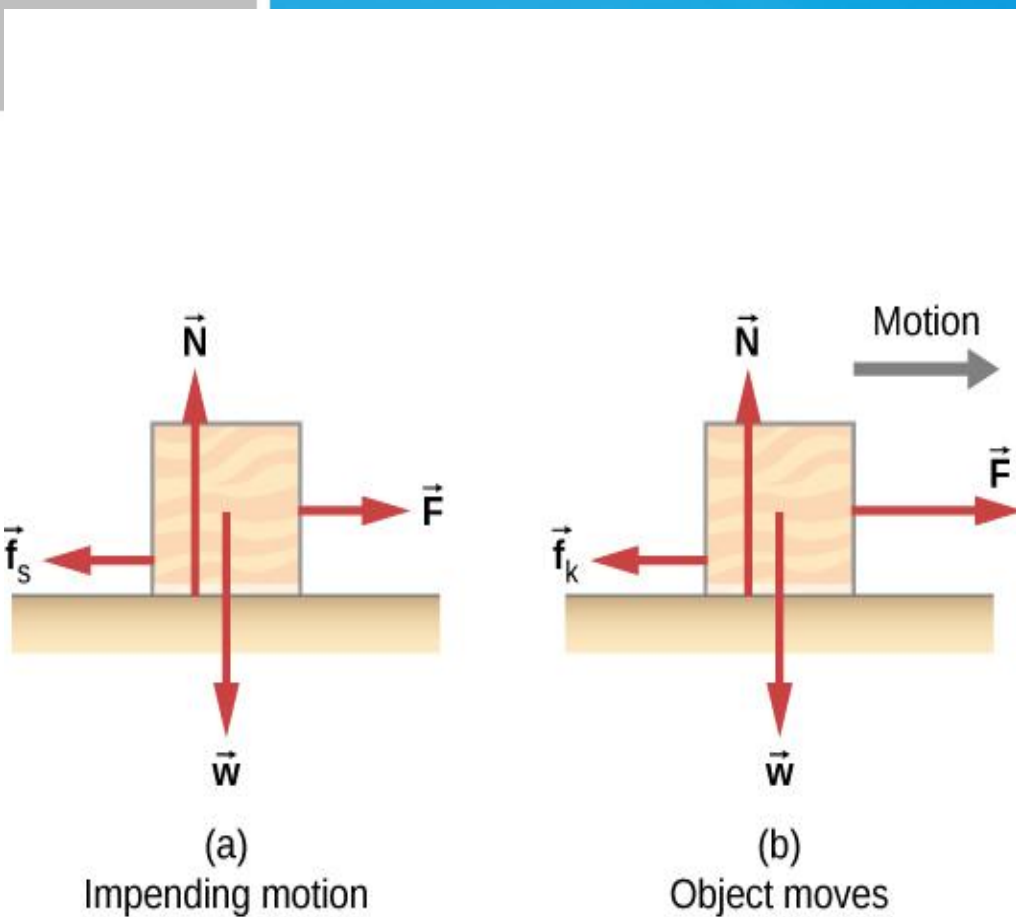


• STATIC FRICTION:

- If two systems are in contact and stationary relative to one another, then the friction between them is called **static friction**.
- The magnitude of static friction f_s is,
- $f_s \leq \mu_s N$,
- where μ_s is the **coefficient of static friction** and N is the magnitude of the normal force.

• KINETIC FRICTION:

- If two systems are in contact and moving relative to one another, then the friction between them is called **kinetic friction**.
- The magnitude of kinetic friction f_k is given by
- $f_k = \mu_k N$,
- where μ_k is the **coefficient of kinetic friction**.



A system in which $f_k = \mu_k N$ is described as a system in which *friction behaves simply*. The transition from static friction to kinetic friction is illustrated in **(Figure)**.

Conservation of Mechanical Energy:

- “In an isolated system in which only conservative forces act the total mechanical energy of the system remains conserved.”

$$K.E_i + P.E_i = K.E_f + P.E_f$$

$$\Delta K.E = -\Delta P.E$$

$$\Delta K.E + \Delta P.E = 0$$



- Examples of conservation of mechanical energy:

- i. Spring Mass system,

- ii. Gravitational field.

Using this definition of the total mechanical energy, Eq. 12-12 becomes

$$\Delta(K_{\text{total}} + U_{\text{total}}) = \Delta E_{\text{total}} = 0. \quad (12-14)$$

For convenience we drop the subscripts “total,” with the understanding that when we apply the result $\Delta(K + U) = \Delta E = 0$ to a system of particles we must always use the *total* values of the various energies for the system.

If the change in any quantity is zero, then that quantity must remain constant, so we can rewrite Eq. 12-14 as

$$E_i = E_f \quad \text{or} \quad K_i + U_i = K_f + U_f, \quad (12-15)$$

where the subscripts *i* and *f* refer to the initial and final states of the system. That is, the initial value of the total mechanical energy is equal to the final value.

Equation 12-15 is the mathematical statement of the *law of conservation of mechanical energy*:

In an isolated system in which only conservative forces act, the total mechanical energy remains constant.

Forces that act within our system can change kinetic into potential or potential into kinetic energy, or even one form of potential energy into another, but the total mechanical energy remains constant. If nonconservative forces, such as friction, act in the system, then the total mechanical energy is *not* constant; we consider this case in Chapter 13.

As an example of the conservation of mechanical energy, let us consider again the block–spring system of Fig.

12-1. At an arbitrary point in the motion, the spring is extended or compressed by a distance x (relative to the $x = 0$ reference position) and the block is moving with speed v , so that the total mechanical energy is $E = \frac{1}{2}mv^2 + \frac{1}{2}kx^2$. When the spring has its maximum extension or compression x_m , the block is instantaneously at rest; at that point the mechanical energy is all potential and $E = \frac{1}{2}kx_m^2$. As the spring then returns to its relaxed length and the block moves toward $x = 0$, the potential energy decreases and the kinetic energy increases, until at $x = 0$ the potential energy becomes zero and the kinetic energy reaches its maximum value of $\frac{1}{2}mv_m^2$, so $E = \frac{1}{2}mv_m^2$. Figure 12-5 illustrates the variation of the kinetic and potential energies as the system moves. Note that at every stage of the motion, the sum $K + U$ remains constant.

Similarly, as the ball at first rises in the ball–Earth system of Fig. 12-2, the gravitational potential energy increases as the kinetic energy decreases, but the total mechanical energy remains constant. Taking $U = 0$ at the point of release, the initial mechanical energy is $E = \frac{1}{2}mv_0^2$. (As we discussed above, the motion of the Earth is negligible in this system, so we can associate the kinetic energy entirely with the ball.) At an arbitrary height y , the total mechanical energy is the sum of the kinetic plus potential energies, $E = \frac{1}{2}mv^2 + mgy$. At its maximum height h , the speed is zero and so $E = mgh$. At each location, the total mechanical energy has the same value, although it may be shared differently between its kinetic and potential parts. As the ball falls,

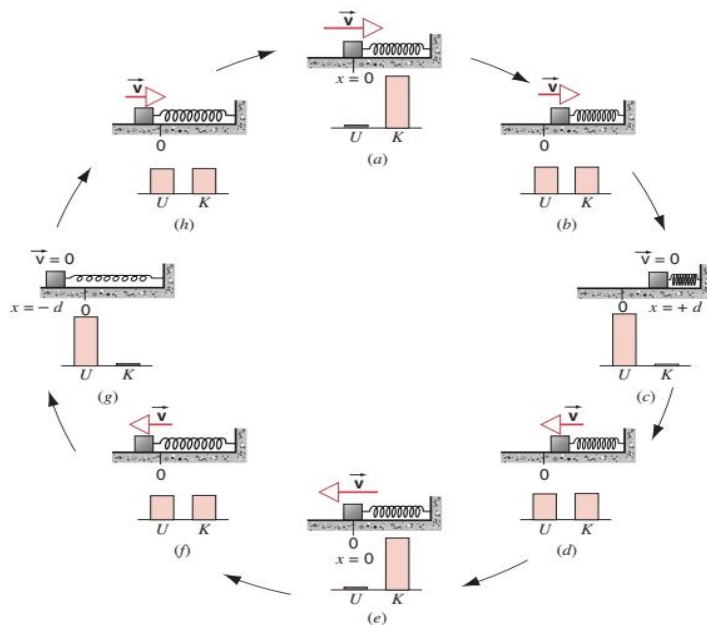


FIGURE 12-5. A block attached to a spring oscillates back and forth on a horizontal frictionless surface. The mechanical energy E of the system remains constant but is shared differently between kinetic and potential energy as the system moves. At certain times (*a*, *e*) the energy is all kinetic, at others (*c*, *g*) it is all potential, and at still others (*b*, *d*, *f*, *h*) it is shared equally between the two forms.

PROBLEM:

A ball of mass " m " drop from a height ' h ' above the ground (neglecting air resistance).

(a) Determine the speed of the ball when it is at a height " y " above the ground?

(b) when it is on the ground?

UNIFORM CIRCULAR MOTION:

- When an object travels in a circular path with constant velocity.
- Characteristics of UCM:
 - Constant velocity
 - constant radius
 - Variable acceleration
 - variable direction
- Centripetal and centrifugal forces are associated with UCM.